

Final System Report

Analytic Approach

Target Definition

The System builds upon the baseline goal of generating financial insights and recommendations, with focus on faithfulness. Specifically, it aims to provide accurate, context-rich advice by leveraging both a domain-fine-tuned Falcon 7B model and advanced prompt optimization techniques.

Inputs

- User Query
 - A text prompt submitted by investors or advisors, potentially vague or incomplete.
 - These queries are refined through an optimized Chain-of-Thought (CoT) before inference.
- Knowledge Base (RAG + Qdrant Vector Database)
 - The same RAG-based knowledge repository used in the baseline.
 - Relevant documents—collected from various sources (e.g., Alpaca) and stored in the Qdrant vector database—are fetched based on query relevance.

What Kind of System Was Built?

A prompt-optimized advisory system that:

- Retains Falcon 7B as the underlying language model (with domain-specific fine-tuning, consistent with the baseline).
- Maintains the baseline's RAG workflow for retrieving relevant information from the Qdrant database.
- **Incorporates GPT based Chain of thought (CoT) program optimized through dspy's MIPROv2 to enhance faithfulness and context alignment.**

Solution Description & Architecture

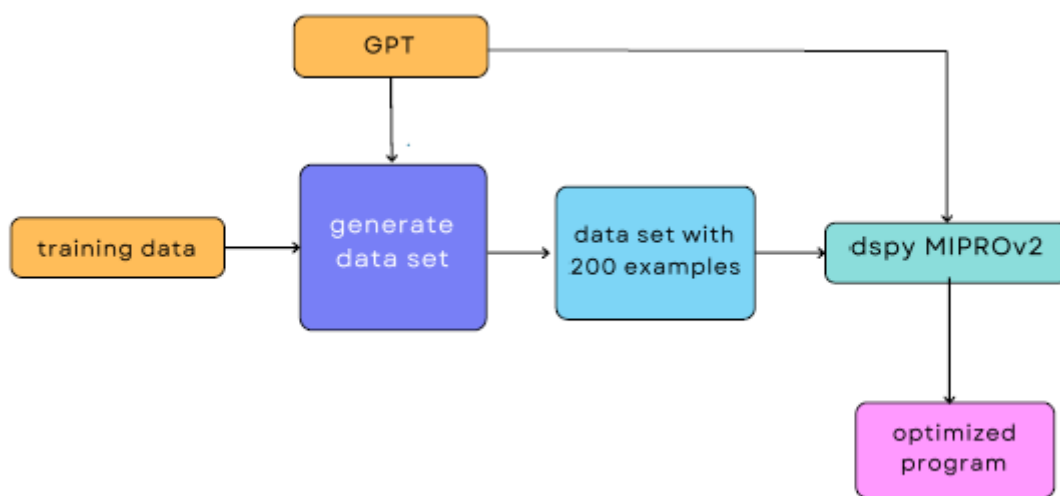
Simple Solution Architecture

- Data Sources
 - User Queries from various investor profiles.
 - Qdrant Vector DB storing relevant financial documents.
- Solution Components
 - Prompt Optimizer (MIPROv2 based CoT program) - Refines user queries and extracted context into a structured Chain-of-Thought. Tested in multiple configurations (Light, Medium, Heavy) for best faithfulness.
- Falcon 7B Model
 - Domain-fine-tuned, as in the baseline.
 - Receives the optimized prompt to generate final answers.
- RAG Layer
 - Same retrieval process as baseline, where relevant documents from Qdrant are appended to the prompt.

Data Flow

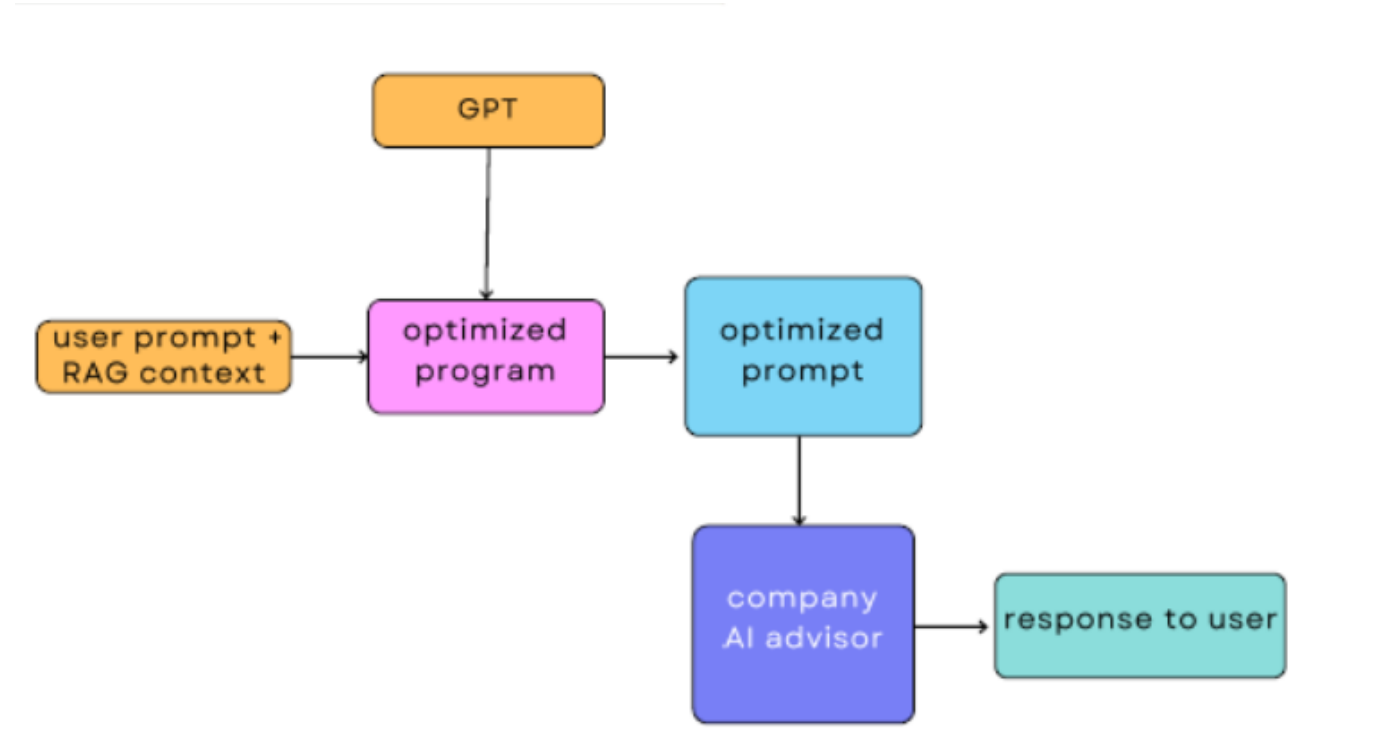
Pre-Deployment

- **Data Generation** - Data samples generated by LLM based on fewshot from the internal dataset examples (randomly chosen). Made in order to collect enough samples in cases of small internal dataset. Having a large, varied set of queries and answers mitigates the risk of MIPROv2 focusing too narrowly on any single question type or domain segment. This ensures the optimized prompts perform robustly across a wide spectrum of financial scenarios.
- **Build CoT optimized program** - The dataset (finetune dataset & synthetic) is used alongside GPT model to construct and optimize a Chain of Thought (CoT) program. The MIPROv2 algorithm, with faithfulness as its key metric, refines these reasoning steps to ensure the model's outputs accurately align with user prompts. Once optimized, the CoT program is stored and made available for seamless integration in production workflows, eliminating the need for repeated large-scale model fine-tuning.



Post-Deployment

- **Initial User Query** - The process begins when a user submits a question or query to the system.
- **Document Retrieval Phase** - The query is sent to a RAG (Retrieval Augmented Generation) system, Qdrant Vector Database is used to store and retrieve relevant documents, The system searches for and retrieves the most relevant documentation based on the query.
- **Prompt Optimization** - The retrieved documents and original query are processed by the optimized CoT program. This component optimizes how the information will be presented to the Falcon language model, it's re-structures the prompt to get the best possible response by automatically adding a query-specific reasoning techniques and fewshot examples.
- **Model Processing & Response Generation** - The optimized prompt is sent to Falcon 7B, a large language model, Falcon 7B processes the prompt along with the retrieved context, the model performs inference to generate a response.



This architecture represents a modern RAG system that combines document retrieval, automatic prompt engineering, and language model capabilities to provide accurate and contextually relevant answers.

Results

MIPROv2 Configurations & Faithfulness

Pre-deployment testing of the MIPROv2-optimized CoT program was conducted across three configurations: Light, Medium, and Heavy. Each configuration was evaluated based on faithfulness scores, with the optimized approach consistently outperforming the baseline. Output was CoT program that was used to optimize the prompt for the Falcon 7B model.

Configuration	Baseline	Avg Minibatch Score	Optimized	Notes
Light	68.8%	66.66%	71.6%	Initial improvement
Medium	68.8%	74.96%	75.18%	Significant gains
Heavy	68.4%	75.18%	76.08%	Highest performance

Across all configurations, the optimized approach outperforms baseline faithfulness scores.

Data Enrichment Example

Based on the MIPROv2-optimized CoT program, the system enriches user queries with additional context and reasoning. This example illustrates how the system enhances user queries to provide more informed and accurate responses.

Original Input Example:

```
{
  "about_me": "I'm a 40-year-old entrepreneur interested in renewable energy.",
```

```
"context": "Solar and wind energy projects are growing globally.",
"question": "Should I invest in renewable energy companies?"
}
```

Enhanced Input Example:

```
{
  "about_me": "I'm a 40-year-old entrepreneur interested in renewable energy.",
  "context": "Solar and wind energy projects are growing globally. Recent trends show increasing government incentives and rising market demand for renewable solutions. Reasoning: Investments in renewable energy companies are supported by global sustainability goals and advancements in solar and wind technologies. These factors make renewable energy an attractive long-term sector for growth.",
  "question": "Should I invest in renewable energy companies?\nYou can use the following expert reasoning as a reference: 'Renewable energy companies are experiencing significant growth due to increased government support and global sustainability goals. Long-term investments in this sector are considered promising.'"
}
```

Comparison of Key Metrics

Compared to the baseline system, the final modified system demonstrates notable improvements:

Metric	Baseline Score	Final System Score	Improvement	Description
Context Relevancy	0.1203	0.1239	+3.0% (approx)	Measures how well the answers align with user queries and relevant documents
Context Recall	0.1343	0.1543	+14.9% (approx)	Evaluates the comprehensiveness of using Qdrant-based data
Faithfulness	0.3365	0.484	+43.8% (approx)	Assesses the accuracy and reliability of outputs based on provided prompts

- **Improved Context Relevancy** - Our solution ensures that responses are closely aligned with the user’s query by better integrating relevant context into the output.
- **Enhanced Context Recall** - Improvement in retrieving and incorporating essential information from the database, resulting in more comprehensive and informed answers.
- **Boost in Faithfulness** - The most notable improvement is in faithfulness, ensuring that outputs are accurate, trustworthy, and closely aligned with the provided input and reasoning.

Final Note

By integrating MIPROv2 prompt optimization and structured Chain-of-Thought reasoning on top of the existing baseline (domain-fine-tuned Falcon 7B + RAG from Qdrant), the final system addresses the key

limitations of vague queries and inconsistent accuracy. The result is a scalable, trustworthy, and higher-performing solution that more effectively aligns with the user's financial needs and context.